Wind-Energy in India

Creating Wind-Energy Feasibility Dashboard

Problem Statement

India's wind energy sector, while a global leader, struggles with three main issues:

- grid instability due to the intermittent nature of wind,
- high costs and difficulty in acquiring land for new projects,
- and a lack of proper infrastructure for offshore wind development. These challenges are slowing down progress towards the country's clean energy goals.

Research Insights

What is the existing scenario ..?

→Wind energy in India is a rapidly growing renewable resource, with India ranking 4th globally in installed capacity. Major wind farms are located in Tamil Nadu, Gujarat, and Maharashtra. Government initiatives, like the Wind-Solar Hybrid Policy, support expansion. Wind energy plays a crucial role in India's sustainable development goals.

Proposed Solution

What are you building (Concept Summary)

→We are building a dashboard on Wind Energy to understand the useful insight to facilitate to solution.

Key Features

Key Features of Wind Energy in India:

India is the 4th Largest wind power producer globally

- **High Potential:** 695 GW at 120m hub height
- **Major States:** Tamil Nadu, Gujarat, Maharashtra
- **Clean & Renewable:** Zero emissions
- **Large Projects:** Muppandal, Jaisalmer Wind Parks
- **Hybrid Focus:** Wind-Solar integration policy
- **Offshore Plans:** Pilot projects in Gujarat, Tamil Nadu
- **Falling Costs:** Competitive with conventional power

Impact / Benefits

1. Change / Impact Brought:

- For Users (Planners, Policymakers): Better decision-making with real-time data
- For Businesses (Investors, Developers): Identify high-potential zones, reduce risk
- For Communities: Improved access to clean energy and job opportunities
- For the Environment: Boosts clean energy adoption, reducing carbon emissions

2. Measure of Success:

- Increased wind project proposals in untapped areas
- Improved efficiency in policy planning and resource allocation
- Higher wind energy contribution in India's energy mix
- Positive feedback and adoption by stakeholders

Go-to-Market Strategy

Phase 1: Research & Planning (Q1): Identify user needs, data sources, and key performance indicators and Collect historical and real-time wind energy data

Phase 2: Dashboard Development (Q2): Design interactive visuals: maps, charts, KPIs and Implement user-friendly UI/UX for policymakers and analysts

Phase 3: Pilot Testing & Feedback (Q3): Deploy dashboard for limited users (government & private stakeholders) and Collect feedback to improve data insights and usability

Phase 4: Scale & Integrate (Q4): Connect with national renewable energy databases and Enable forecasting, storage analysis, and hybrid energy tracking

Future Developments: Add **offshore wind energy tracking**, Integrate **AI-based prediction models**, Expand to **global benchmarking & comparative analysis**

Team & Roles

Team Members:

- **MD USMAN ANSARI :**

  Importing Raw Data for Analysis,

  Creating Dashboard and ppt

- **VASAMSETTI GEETHA SANDHYA:**

  Cleaning the data and writing the documentations

THANK YOU